

Immune potentiation of ultrafine dietary particles in normal subjects and patients with inflammatory bowel disease.

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Various specific and non-specific environmental factors have been associated with the induction and/or exacerbation of disease activity in patients with Crohn's disease and ulcerative colitis. One such factor is the potential role of ingested ultrafine particles. In fact, based on a Western diet, recent data suggest that more than 10(12)ultrafine particles are ingested per person every day. These microparticles have been considered inert although they adsorb endogenous constituents of the intestinal lumen and are taken up by human intestinal lymphoid aggregates. Based on these observations, we determined whether one such dietary microparticle, titanium dioxide (TiO₂), alters intestinal cell responsiveness to lipopolysaccharide (LPS) using colonic biopsy specimens from 28 patients with ulcerative colitis, 21 with Crohn's disease, and 36 healthy controls. These samples, as well as peripheral blood mononuclear cells when available, were incubated alone (control), or with either (a) LPS (1-2,000 ng/ml), (b) TiO₂(5 microg/ml) or (c) LPS (1 ng/ml) adsorbed to TiO₂(5 microg/ml). In each case, the levels of interleukin 1 (IL-1) produced in these assays were quantitated by bioassay and by ELISA. Interestingly, there was dramatic stimulation of peripheral blood mononuclear cells using the TiO₂-LPS conjugate, with values 30-60-fold above controls and only minor stimulation with LPS or TiO₂alone. In intestinal organ cultures there was no increase in IL-1 secretion when challenged with TiO₂alone or with up to 2,000 ng/ml LPS. However, the TiO₂-LPS conjugate produced a two-to-three-fold, significant increase in the intestinal secretion of IL-1. Our data demonstrate that ultrafine dietary particles are not immunologically inert and may be important adjuncts in overcoming normal gut cell hyporesponsiveness to endogenous luminal molecules. This may be particularly relevant to patients with inflammatory bowel disease where there is abnormal intestinal permeability. Copyright 2000 Academic Press.